Do Now · retrieval

Final specification and synoptic checkpoint

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Name a PageRank property beyond counting inbound links.

Q2

What is a page fault?

Q3

What does an INNER JOIN return?

Q4

Search value 11 in [1,3,5,7,9,11,13], showing mid indexes.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Use evidence rather than ticks alone — A topic is secure when a learner can solve an unfamiliar example, not merely recognise its name.

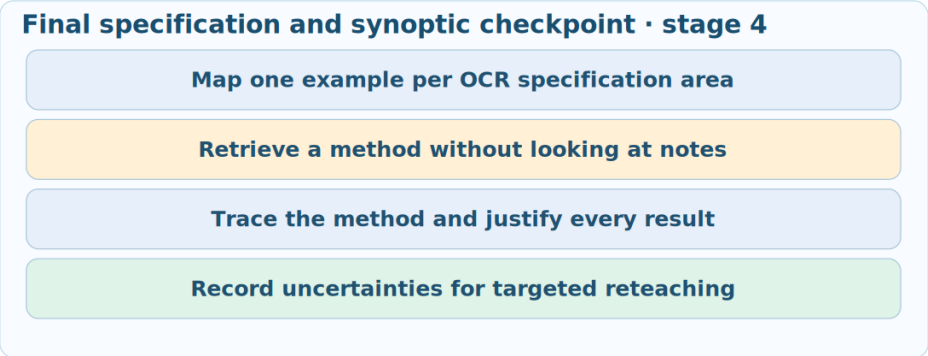
Core idea 2

Find links between papers — Paper 1 numerical representation and database knowledge can underpin Paper 2 programming and algorithm questions.

Core idea 3

Prioritise gaps — Record the exact misconception or missing step so later Paper 1/2 reteaching sessions target observable weaknesses.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

synoptic — Combining knowledge across topics.

retrieval — Reproduce knowledge from memory.

misconception — Incorrect conceptual understanding.

intervention — Specific teaching targeting a gap.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

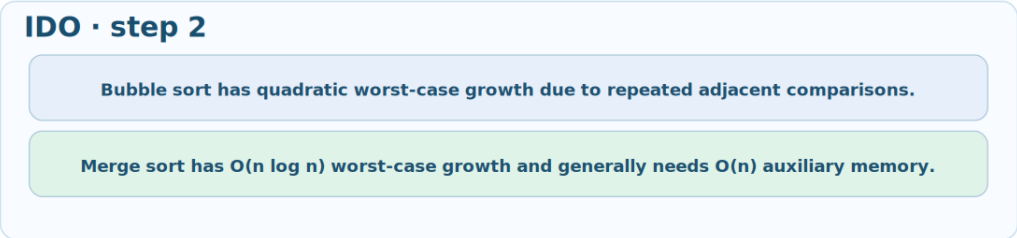
Evaluate the claim 'all sorts take O(n log n) in the worst case'.

Follow the worked method

1. Predict first — Bubble sort has quadratic worst-case growth due to repeated adjacent comparisons.

2. Check reasoning — Merge sort has O(n log n) worst-case growth and generally needs O(n) auxiliary memory.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

Identify exactly which implementation state is wrong, e.g. high never decreases.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

State roles of PC, MAR, MDR, CIR and ACC, plus one bus role. [6]

Question 2

A sorted list [2,4,6,8,10,12,14] is searched for 12. Trace binary search, then explain worst-case Big O. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A school chooses AI scoring for pupil support. Give three contextual safeguards and explain risks. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Knowing a topic name alone proves that you can apply the associated algorithm.

Use key vocabulary from memory

1. Explain synoptic in your own words.

2. Explain retrieval in your own words.

3. Explain misconception in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.